Target E-commerce – Requirements

1. Context

Target is a globally renowned retail brand and one of the largest retailers in the United States. Target differentiates itself by delivering outstanding value, innovation, and an exceptional guest experience.

This business case focuses on **Target's e-commerce operations in Brazil**, analyzing **~100,000 orders placed between 2016 and 2018**.
The dataset provides a comprehensive view of:

- Order lifecycle and delivery performance
- Pricing, freight, and payment behavior
- Customer demographics and geographic distribution
- Product characteristics
- Customer satisfaction through reviews

The objective is to **extract actionable insights** that can support operational optimization, logistics planning, and business strategy decisions.

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2. Dataset & Engineering Scope

Dataset Source

📁 [Target Dataset]

Available Tables (CSV)

- customers
- sellers
- order_items
- geolocation
- payments
- reviews
- orders
- products

Customers

Column Name	Description
customer_id	ID of the consumer who made the purchase
customer_unique_id	Unique identifier of the consumer
customer_zip_code_prefix	ZIP code prefix of the customer’s location
customer_city	City from where the order was placed
customer_state	State code from where the order was placed (e.g., São Paulo – SP)

Geolocation

Column Name	Description
geolocation_zip_code_prefix	First 5 digits of the ZIP code
geolocation_lat	Latitude
geolocation_lng	Longitude
geolocation_city	City
geolocation_state	State

Seller

Column Name	Description
seller_id	Unique identifier of the seller registered
seller_zip_code_prefix	ZIP code prefix of the seller’s location
seller_city	City of the seller
seller_state	State code (e.g., São Paulo – SP)

Order_Items

Column Name	Description
order_id	Unique identifier of the order
order_item_id	Unique identifier for each item within an order
product_id	Unique identifier of the product
seller_id	Unique identifier of the seller
shipping_limit_date	Date before which the product must be shipped
price	Actual price of the product
freight_value	Cost of shipping the product

Payments

Column Name	Description
order_id	Unique identifier of the order
payment_sequential	Sequence number of payments (for EMI cases)
payment_type	Payment method used (e.g., Credit Card, Boleto)
payment_installments	Number of installments for EMI purchases
payment_value	Total amount paid for the order

Orders

Column Name	Description
order_id	Unique identifier of the order
customer_id	Identifier of the customer who placed the order
order_status	Current status of the order (e.g., delivered, shipped)
order_purchase_timestamp	Timestamp when the order was placed
order_delivered_carrier_date	Date when the carrier delivered the order

order_delivered_customer_date	Date when the customer received the order
order_estimated_delivery_date	Estimated delivery date

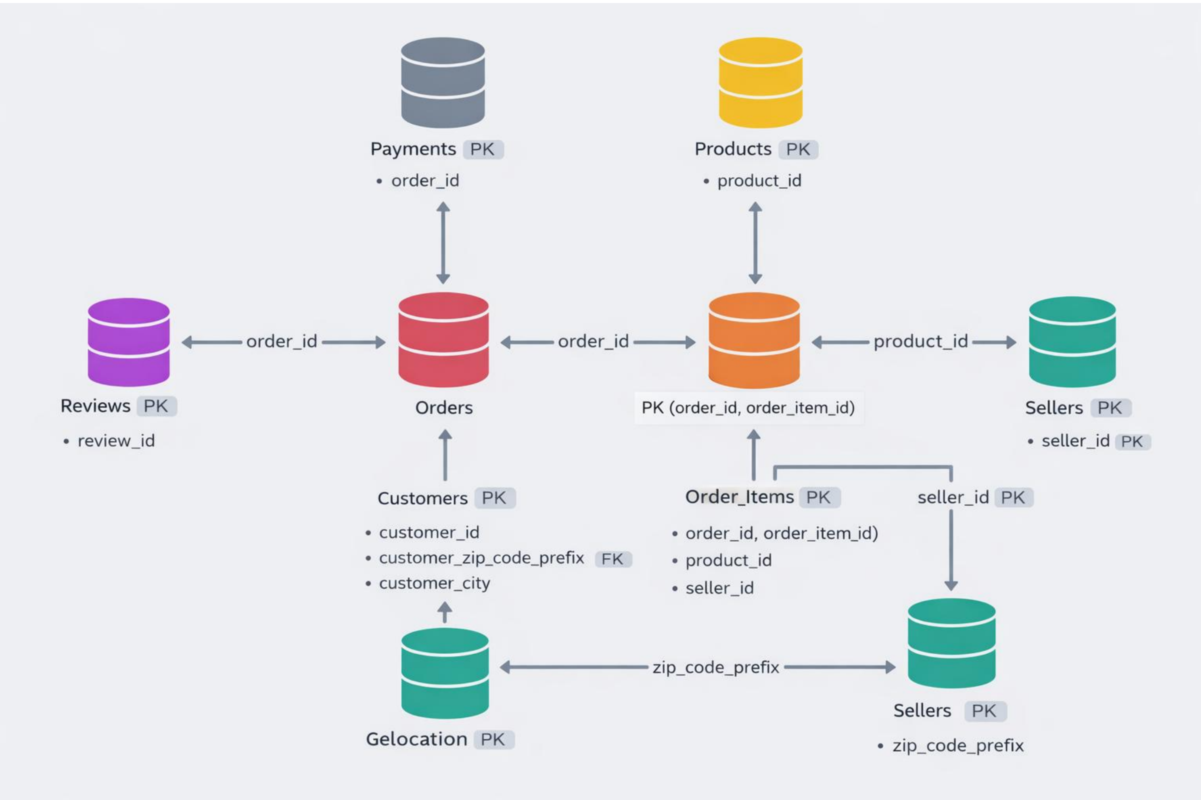
Reviews

Column Name	Description
review_id	Unique identifier of the review
order_id	Unique identifier of the order
review_score	Rating given by the customer (1–5)
review_comment_title	Title of the review
review_comment_message	Review comments provided by the customer
review_creation_date	Timestamp when the review was created
review_answer_timestamp	Timestamp when the review was answered

Products

Column Name	Description
product_id	Unique identifier of the product
product_category_name	Name of the product category
product_name_lenght	Length of the product name string
product_description_lenght	Length of the product description
product_photos_qty	Number of product photos available
product_weight_g	Weight of the product in grams
product_length_cm	Length of the product in centimeters
product_height_cm	Height of the product in centimeters
product_width_cm	Width of the product in centimeters

Dataset schema:



Phase 1: Data Loading & Schema Validation

1. Load all datasets into a relational database.
2. Inspect and document:
   - Column data types
   - Primary keys and foreign-key relationships
3. Validate:
   - Referential integrity between tables
   - Presence of nulls and inconsistent values
4. Identify:
   - Order date range covered by the dataset
   - Number of unique customer cities and states

Phase 2: Order Volume & Time-Based Analysis

1. Build queries to analyze:
   - Year-over-year growth in order volume
   - Monthly order trends and seasonality
2. Implement **time-bucketing logic** to classify order timestamps into:
   - Dawn (00–06)
   - Morning (07–12)
   - Afternoon (13–18)
   - Night (19–23)

Phase 3: Regional & State-Level Analytics

1. Compute **month-on-month order volume per state**.
2. Design queries that:
   - Aggregate customers by state
   - Enable easy extension to city-level analysis
3. Ensure outputs are structured for downstream dashboards or APIs.

Phase 4: Economic Impact & Revenue Analysis

1. Calculate the **percentage increase in total order value** from **2017 to 2018**
   - Limit analysis to **January–August**
   - Use payment_value as the monetary metric
2. Compute:
   - Total and average order value per state
   - Total and average freight cost per state

Phase 5: Delivery Performance & Logistics Metrics

Using a **single optimized query**, compute:

Delivery time (days):

time_to_deliver = order_delivered_customer_date - order_purchase_timestamp

Estimated vs actual delivery difference (days):

diff_estimated_delivery = order_delivered_customer_date - order_estimated_delivery_date

Based on these metrics:

1. Identify:
    - Top 5 states with **highest** average freight cost
    - Top 5 states with **lowest** average freight cost
2. Identify:
    - Top 5 states with **slowest** average delivery
    - Top 5 states with **fastest** average delivery
3. Identify:
    - Top 5 states where deliveries are **significantly faster than estimated**

Phase 6: Payment System Analysis

1. Generate **month-on-month order counts** grouped by:
    - Payment type
2. Analyze order distribution by:
    - Number of payment instalments
